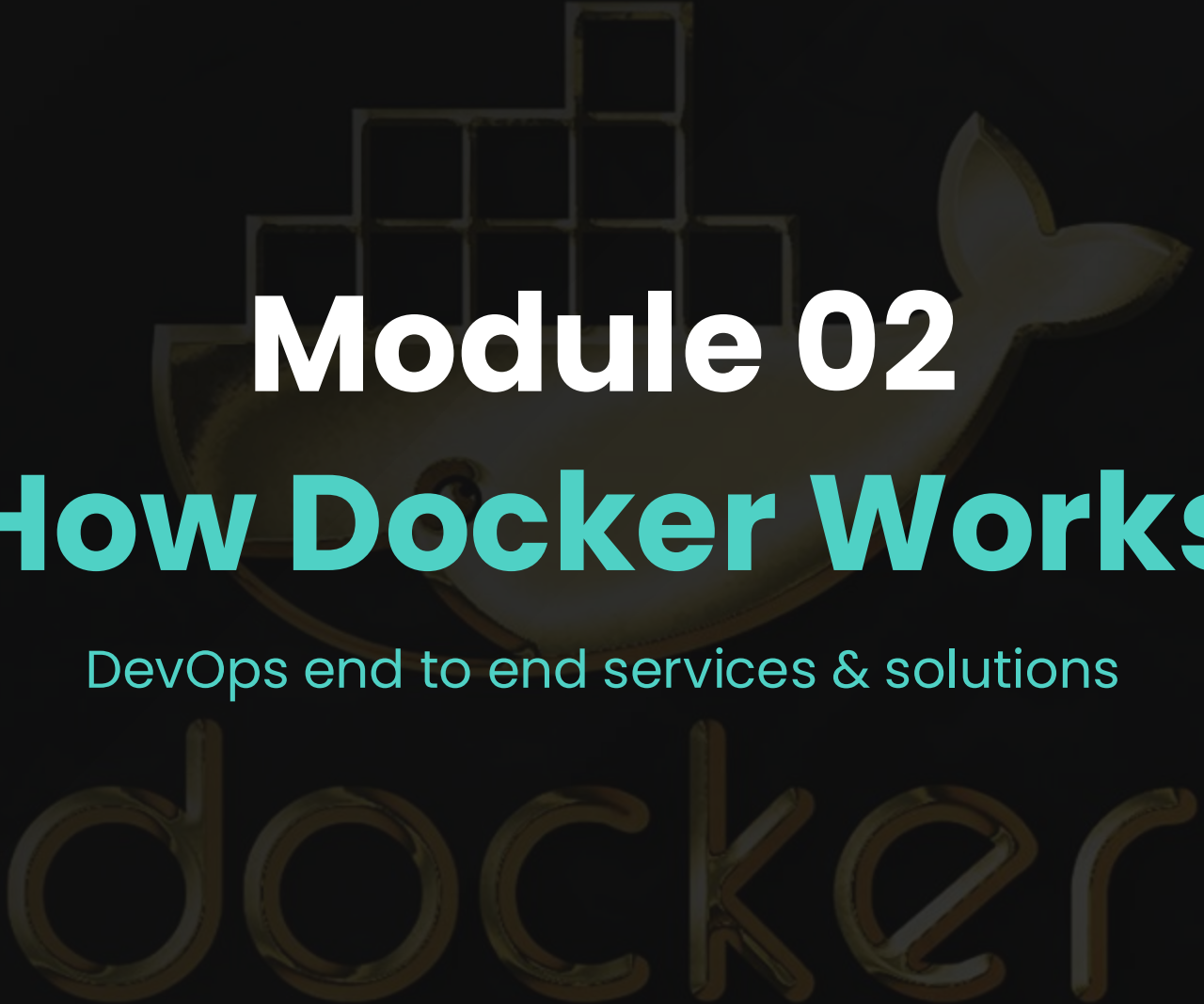




Module 02

How Docker Works

DevOps end to end services & solutions

Agenda

- > How Docker Works
- > Docker vs Virtual Machines
- > Getting Started with Docker
- > Dockerfile Basics
- > Hands-On – Lab-01 – Getting Started with Docker

How Docker Works



Dockerfile

Script to build Docker images.



Build

Create an image from a Dockerfile.



Run

Start a container from an image.



Registry

Store and distribute Docker images.

Docker vs Virtual Machine

Virtual Machines

Require a hypervisor, include a full Guest Operating System, and are generally heavy on resources.

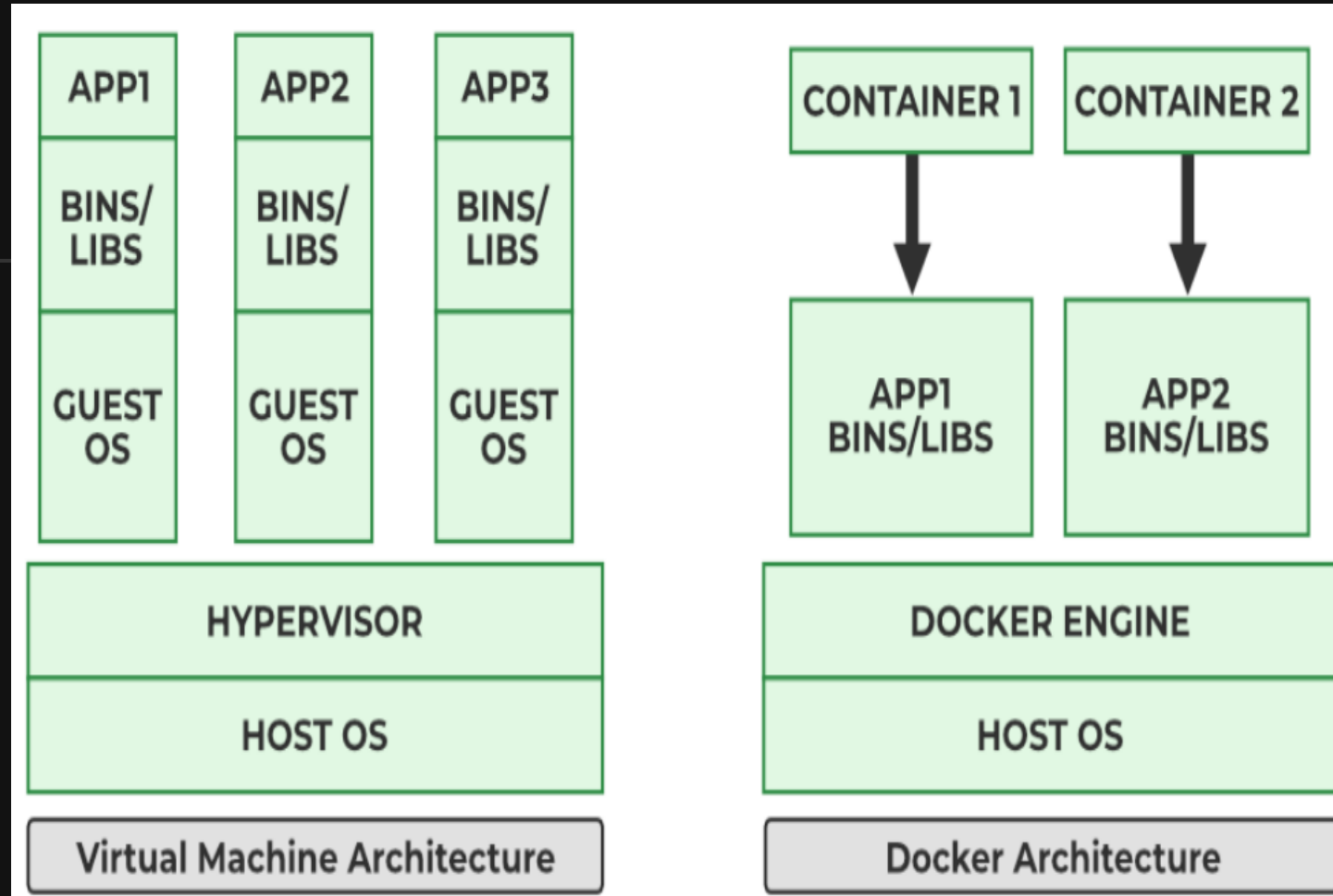
Docker Containers

Share the host OS kernel, are extremely lightweight, and have a very fast startup time.

Architecture Comparison

The visual difference highlights the efficiency of containers.

- > **VMs:** Each app has its own Guest OS.
- > **Containers:** Apps share the Container Engine & Host OS.



Dockerfile Basics



FROM

Base
image



COPY

Copy files from host to
container



RUN

Execute commands in
the container



CMD

Command to run when
container starts

Dockerfile Example

Structure Overview

A standard Dockerfile instructions flow:

- > Start with an OS base.
- > Add your application code.
- > Install dependencies.
- > Define the launch command.

```
FROM ubuntu:latest
COPY . /app
RUN make /app
CMD [ "./app/run.sh" ]
```

Hands-On – Lab-02

- > **Goal:** Getting Started with Docker
- > Docker pull and run the Monopoly game on one command
- > Browse via Google Chrome
- > Display the images
- > Display the container run time and exited

Questions?

Thank you for participating.

Feel free to reach out:
<https://dinghy-e2e.com>

